SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I, Mohammed Ilyas - 192325017 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ilyas

Dudekula Mohammad Ilyas
192325017

3)

Classical Encryption Schemes

Substitution Techniques
(Replace plaintext symbols with other symbols)

Types
- Caeser Cipher
- Monoalphabetic cipher
- Vigenere cipher
- Hill cipher

Effect on Data Structure
- Actual characters are changed with other symbols
- Length of plain text remains same.
- original students/structure is modified
- Ex: HELLO → KHOOR

Security level
- Caesar - Low security
- Monoalphabetic - low security
- vigenere - Higher security (uses multiple alphabets)
- Hill - Higher security (uses matrix multiplication)
- Depends on key and complexity

Transposition Techniques
(Rearrange positions of plaintext symbols)

Types
- Rail Fence cipher
- Columnar Transposition
- Permutation cipher

Effect on Data Structure
- character remains the same but their positions are rearranged
- Only order changes.
- length of plan text remains same.

Ex: HELLO → HLEOL

Security level
- Generally less secure than complex substitution techniques.
- Vulnerable to pattern analysis.
- Security depends on the complexity of rearrangement
- Can be combined with for better security

Comparision of Substitution and Transpotion Techniques

Substitution Techniques

- Changes actual symbols in the plaintext
- Content is modified
- security depends on key and complexity of substitution

Transposition Techniques

- Changes only the positions of symbols
- Content remains same.
- security depends on complexity of rearrangement

Similarities

- Both are classical encryption techniques
- Both make plain text unreadable
- Both use reverse length of plaintext

Differences

Aspect	substitution	Transposition
What changes	Actual symbols	positions of symbols
Data Structure	modified	Remains same
Method	Symbol replacement	Rearrange ment.

vulnerability to attacks

- Simple substitution is vulnerable to frequency analysis.
- Transposition is vulnerable to pattern analysis and brute-force attacks.

5)

Number Theory concepts in cryptography

Main Concepts

Modular Inverse
- finds a^{-1} such that $a \times a^{-1} = 1$
- used to find decryption key

Extended Euclid algorithm
- finds gcd(a,b) & co-efficients x,y such that $ax+by = gcd(a,b)$
- used to compute

Fermat's little
- If p is prime number $a^{p-1} = 1 \pmod p$
- use in modular arithmetic

Euler's phi function
- $\psi(n)$ = number of integers ($1 \le k < n$) that are coprime to n.

for RSA:
$\psi(n) = (p-1)(q-1)$

Mathematical Connections

- Extended Euclid Algorithm is used to find mod Inverse
- Fermat's little Theorm is a special case of euler's theorm
- Euler phi function is used in Euler's Theorm
- Modular Inverse, Euler's phi function and fermat's little theorm are all based on modular arithmetic

Role in Encryption / Description.

modular Inverse
- used to find description key e (or d) in RSA
- solves equations like $ed \equiv 1 \pmod{\Phi(n)}$.

Extended Euclid
- used to compute modular inverse
- finds x,y such that $ax+by=1$

fermat's Little Theorm.
- used in encryption / description with prime modules.
- Basics for many crypt algorithms.

Euler's Theorm
- Used to prove the correctness of RSA
- $a^{\Phi(n)} = 1 \pmod n$ helps in decryption

Relationships B/w Concepts

Euclid Algorithm

- Finds gcd(a,b) and x,y such that $ax + by = gcd(a,b)$
- used to find mod Inverse

→ **modular Inverse**

$a \times a^{-1} = 1 \pmod m$

Used to calculate decryption key in RSA

fermat's Little Theorm

If p is prime

$a^{p-1} = 1 \pmod p$

Special case of Euler's Theorm

Euler's phi func

- $\phi(n)$ = no of Integers less than that are co-prime to n.
- for RSA: $\phi(n) = (p-1)(q-1)$

→

Eulers theorm

① If gcd(a,n) = 1

$a^{\phi(n)} = 1 \pmod n$

② uses $\phi(n)$

Flow in RSA using NO theory concepts.

Choose two large primes p,q → Compute $n = p \times q$ → Compute $\phi(n) = (p-1)(q-1)$ → Choose e such that gcd(e,4) = 1 → Use euclid Algo to find d.

- Public Key (e,n) (Encryption)
- Private Key (d,n) (Decryption.)